types of caveats that it supports. Trying to add a caveat that the API doesn’t support will prevent the macaroon from being used. Update the `read` method to verify the new caveat as shown in the listing.

Listing 9.17 Verifying the expiry timestamp

```
@Override
public Optional<Token> read(Request request, String tokenId)
{
var macaroon = MacaroonsBuilder.deserialize(tokenId);
var verifier = new MacaroonsVerifier(macaroon);
verifier.satisfyGeneral(new TimestampCaveatVerifier()); #A
if (verifier.isValid(macKey.getEncoded())) {
return delegate.read(request, macaroon.identifier);
}
return Optional.empty();
}
```

#A Add a TimestampCaveatVerifier to satisfy the expiry caveat.

You can also add your own caveat verifiers using two methods. The simplest is the `satisfyExact` method, which will satisfy caveats that exactly match the given string. For example, you can allow a client to restrict a macaroon to a single type of HTTP method by adding the line

```
verifier.satisfyExact("method = " + request.requestMethod());
```

to the read method. This ensures that a macaroon with the caveat `method = GET` can only be used on HTTP GET requests, effectively making it read-only. A more general approach is